# Solving Numerical Problems - Mark scheme

Number · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

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## Question 1

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2025 · Question 11*

| Q | Working | Answer | Mark | Notes |
|---|---|---|---|---|
| | eg (1 box =) 708 ÷ 12 (= 59)<br>**or** (2 boxes =) 708 ÷ 6 (= 118)<br>**or** (3 boxes =) 708 ÷ 4 (= 177)<br>**or** (4 boxes =) 708 ÷ 3 (= 236)<br>**or** (6 boxes =) 708 ÷ 2 (= 354)<br><br>**or** 12 ÷ 5 (= 2.4) **or** $708 \div \frac{12}{5}$ oe<br><br>eg 708 ÷ 2.4 **or** $708 \times \frac{5}{12}$<br><br>**or** 708 × 5 (3540) | | 2 | M1 for starting to work with proportion |
| | *Correct answer scores full marks (unless from obvious incorrect working)* | 295 | | A1 cao |
| | | | | **Total 2 marks** |

## Question 2

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2021 · Question 6*

| Q | Working | Answer | Mark | Notes |
|---|---|---|---|---|
| | 6 × 100 (= 600) **or** 17.5 ÷ 100 (= 0.175) | | 3 | B1 |
| | "600" ÷ 17.5 (= 34.28…) **or** 6 ÷ "0.175" (= 34.28…) | | | M1 ft incorrect conversion |
| | | 34 | | A1 cao |
| | | | | **Total 3 marks** |

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## Question 3

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2F · November 2023 · Question 15*

| Question | Working | Answer | Mark | Notes | |
|---|---|---|---|---|---|
| | **SMALL**<br>(£ per g) $\frac{1.80}{1.50}(=0.012)$ or (p per g) $\frac{180}{150}(=1.2)$ or<br>(g per £) $\frac{150}{1.80}(=83.(3...))$ or (g per p) $\frac{1.50}{1.80}(=0.83..)$<br><br>**LARGE**<br>(£ per g) $\frac{5}{400}(=0.0125)$ or (p per g) $\frac{500}{400}(=1.25)$<br>(g per £) $\frac{400}{5}(=80)$ or (g per p) $\frac{400}{500}(=0.8)$<br>**COMPARISON OF SAME AMOUNTS EXAMPLE**<br>eg cost of 1200 g = $\frac{1200}{150}\times 1.80(=14.4)$ or | | 3 | M1 A start to comparing costs by finding cost per gram for either size bag or for finding grams per pound (or pence) for either size bag or for finding the cost of a weight of fudge for either size bag. | M2 for a fully correct method to reach a conclusion eg $\frac{400}{150}\times 1.80(=4.8)$ or $\frac{150}{400}\times 5(=1.875)$ or $\frac{5}{1.80}\times 150$ (=416.66…) or $\frac{1.80}{5}\times 400(=144)$ or<br>300 g = £3.60<br>100 g = £1.20<br>400 g = £4.80<br><br>etc |
| | cost of 1200 g = $\frac{1200}{400}\times 5(=15)$ oe | | | | |
| | | | | M1 2 correct compatible calculations | |
| | *Working required* | Small | | A1 Correct conclusion with correct values from correct working dep on M2 | |
| | | | | | **Total 3 marks** |

## Question 4

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2025 · Question 1*

| Q | Working | Answer | Mark | Notes |
|---|---|---|---|---|
| (a) | | 725 | 1 | B1 cao |
| (b) | | 32 | 1 | B1 cao |
| (c) (i) | | 4578 | 1 | B1 cao |
| (ii) | | 8754 | 1 | B1 cao |
| | | | | **Total 4 marks** |

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## Question 5

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2025 · Question 7*

| Q | Working | Answer | Mark | Notes |
|---|---|---|---|---|
| | $\frac{1}{4}\times 26.8\,(=6.7)$ oe **or** $\frac{3}{4}\times 26.8\,(=20.1)$ oe<br>**or** $2\times 26.8\ (=53.6)$ | | 4 | M1 for method to find the discount off one calculator **or** the reduced price of one calculator **or** the cost of two calculators |
| | "6.7" × 3 × 2 (= 40.2) oe eg (26.8 – "6.7") × 2 (= 40.2)<br>**or** "20.1" × 2 (= 40.2) **or**<br>$"53.6"-\frac{1}{4}\times "53.6"\,(=40.2)$ oe<br>**or** $\frac{3}{4}\times "53.6"\,(=40.2)$ oe | | | M1 for method to find the reduced price of two calculators |
| | 50 – "40.2" | | | M1 for a complete method |
| | *Correct answer scores full marks (unless from obvious incorrect working)* | 9.8(0) | | A1<br>SCB2 for an answer of 29.9(0) **or** 36.6(0)<br>SCB1 for an answer of 23.2(0) **or** ±3.1(0) |
| | | | | **Total 4 marks** |

## Question 6

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2022 · Question 13*

| Q | Working | Answer | Mark | Notes | |
|---|---|---|---|---|---|
| | eg 2.5 × 6.5 (= 16.25)<br>**or** 0.5 × 6.5 × 1 (= 3.25)<br>**or** 3.5 × 6.5 (= 22.75) | | 4 | M1 | M2 for 0.5(2.5 + 3.5) × 6.5 (= 19.5) **or** 2 × (0.5(2.5 + 3.5) × 6.5) (= 39) |
| | 2.5 × 6.5 + 0.5 × 6.5 × 1 (= 19.5)<br>**or** 2 × (2.5 × 6.5 + 0.5 × 6.5 × 1) (= 39)<br>**or** 3.5 × 6.5 – 0.5 × 6.5 × 1 (= 19.5)<br>**or** 2 × (3.5 × 6.5 – 0.5 × 6.5 × 1) (= 39) | | | M1 | |
| | 2 × "19.5" ÷ 12 (= 3.25)<br>**or** "39" ÷ 12 (= 3.25)<br>**or** 12 + 12 + 12 + 12 (= 48)<br>**or** 4 × 12 (= 48) | | | M1 | **or** [their area] ÷ 12 (dep on M1)<br>**or** using multiples of 12 for [their area] eg area = 19.5 **and** 12 + 12 (= 24) **or** 2 × 12 (= 24) |
| | | 4 | | A1 | dep on M2, must be from correct working |
| | | | | | **Total 4 marks** |

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## CHALLENGE QUESTIONS

### Question 7

*Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2013 · Question 9*

| | | | | |
|---|---|---|---|---|
| 9 | **Alternative - with approximation**<br>$\frac{x}{y}$ or $x \times \frac{1}{y}$<br>where $x$ is $35\,000 \leq x \leq 40\,000$<br><br>**AND**<br><br>$336 \leq y \leq 400$ | | 4 | M2<br>M1 for $\frac{x}{y}$ or $x \times \frac{1}{y}$ where either the value of $x$ **or** the value of $y$ is acceptable |
| | | integer in the range 93 – 111 inclusive | | **The award of any accuracy marks is dependent on the award of M2**<br><br>A2 A1 for non-integer in the range 93 – 111 |
| | | | | **Total 4 marks** |

### Question 8

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2023 · Question 6*

| | | | | |
|---|---|---|---|---|
| 6 | eg<br>$1 - \frac{1}{6}$ or $\frac{5}{6}$<br>1 – 0.16(666…) (= 0.83(333…) oe or<br>100(%) – 16(.666…)(%) (= 83(.333…)(%)) oe<br>or<br>1 – 0.2 (= 0.8) oe or<br>100(%) – 20(%) (= 80(%)) | | 4 | M1<br>Allow eg<br>1 – 0.16 (= 0.84) (= 84(%))<br>1 – 0.17 (= 0.83) (= 83(%))<br>rounded or truncated |
| | 140 , "$\frac{5}{6}$" (= 168) oe or<br>140 ÷ "0.83(333…)" (= 168) oe eg 140 ÷ 83.33 × 100 (=168)<br>**or**<br>136 ÷ "0.8" (= 170) oe eg 136 ÷ 80 × 100 (= 170) | | | M1<br>Allow eg<br>140 ÷ "0.84" = 166(.666…)<br>140 ÷ "0.83" = 168(.674…)<br>rounded or truncated |
| | 140 , "$\frac{5}{6}$" (= 168) oe or<br>140 ÷ "0.83(333…)" (= 168) oe<br>**and**<br>136 ÷ "0.8" (= 170) oe | | | M1<br>Allow eg<br>140 ÷ "0.84" = 166(.666…)<br>140 ÷ "0.83" = 168(.674…)<br>rounded or truncated |
| | *Working not required, so correct answer scores full marks (unless from obvious incorrect working)* | 2 | | A1 Allow –2 |
| | | | | **Total 4 marks** |

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## Question 9

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2025 · Question 4*

| | | | | | |
|---|---|---|---|---|---|
| 4 | $45 \div 1.5\ (=30)$ **or** $5 \times 12\ (=60)$ **or** $\frac{64}{1.5}\ \left(=\frac{128}{3}=42.6(6...)\right)$ | | 5 | M1 | for a method to find the number of boxes needed **or** the cost of adhesive **or** cost of tiles per $m^2$ |
| | "30" × 64 (= 1920) **or** "42.6(6…)" × 45 (=1920) | | | M1 | for a method to find the cost of the boxes of tiles |
| | "1920" + "60" (= 1980) **or** 3000 – 1920 – 60 (=1020) | | | M1 | for a method to find the total cost **or** the profit |
| | eg $\frac{3000-"1980"}{"1980"}\ (=0.515...)$ **or** $\frac{3000-"1980"}{"1980"}\times 100$ **or** $\frac{3000}{"1980"}\ (=1.515...)$ **or** $\frac{3000}{"1980"}'\ 100\ (=151.5...)$ **or** $\frac{3000}{"1980"}\times 100-100$ | | | M1 | for a method to find the percentage profit or be one step away |
| | *Correct answer scores full marks (unless from obvious incorrect working)* | 51.5 | | A1 | awrt 51.5<br><br>SCB3 for answer 56.3 or 56.25 (from use of 1920 instead of 1980 as total cost) |
| | | | | | **Total 5 marks** |

## Question 10

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2022 · Question 13*

| | | | | |
|---|---|---|---|---|
| 13 | $\frac{3}{8}+45\%\ \left(=\frac{33}{40}\text{ or }82.5(\%)\text{or }0.825\right)$ | | 5 | M1 Do **NOT** award M1 for e.g. $\frac{3}{8}+45(\%)+406(=....)$ oe |
| | $1-\frac{"33"}{40}\left(=\frac{7}{40}\right)$ **or** $100-"82.5"(\%)\left(=17.5(\%)\right)$ **or** $1-"0.825"(=0.175)$ | | | M1 |
| | $406 \div \frac{"7"}{40}\ (=2320)$ **or** $406 \div \frac{"17.5"}{100}$ oe $(=2320)$ **or** 1% = 406 ÷ "17.5" (= 23.2) oe | | | M1 |
| | 0.45 × "2320" oe **or** 45 × "23.2" oe | | | M1 |
| | | 1044 | | A1 |
| | | | | **Total 5 marks** |

| | | | | |
|---|---|---|---|---|
| 13 ALT | $\frac{3}{8}x+0.45x+406$ oe | | 5 | M1 Do **NOT** award M1 for e.g. $\frac{3}{8}+45(\%)+406(=....)$ oe |
| | $\frac{3}{8}x+0.45x+406=x$ oe | | | M1 for a correct equation |
| | $(x=)\frac{406}{1-\frac{3}{8}-0.45}\left(=\frac{406}{7/40}=2320\right)$ | | | M1 |
| | 0.45 × "2320" | | | M1 |
| | | 1044 | | A1 |
| | | | | **Total 5 marks** |

## Question 11

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2024 · Question 14*

| | | | | | |
|---|---|---|---|---|---|
| **14** | $1 + 1.45\ (= 2.45)$ or $1 + \frac{29}{20}\ (= \frac{49}{20})$ or $B = 1.45A$ oe or $B = \frac{29}{20}A$ oe or<br>$A + 1.45A$ or $A + \frac{29}{20}A$ or $2.45A$ or<br>$(A : B =)\ 100 : 145$ oe or $100 + 145\ (= 245)$ oe or<br>$(B : C =)\ 3 : 2$ oe or $B = 1.5C$ oe | | | 5 | B1 must identify ratios with Abel and Bahira or Bahira and Chanda<br><br>Allow any letters for A, B and C |
| | $A + 1.45A = 15\,435$ or<br>$15\,435 \div$ "2.45" or $15\,435 \div$ "$\frac{49}{20}$" or<br>$15\,435 \div$ "245" $\times 100$ or $63 \times 100\ (= 6300)$ | M2 for<br>$15\,435 \div \left(\frac{1}{1.45} + 1\right)\ (= 9135)$ oe or<br>$15\,435 \div \left(\frac{49}{29}\right)\ (= 9135)$ oe | | | M1 for a method to find Abel's savings<br>or<br>for 6300 |
| | $15\,435 -$ "6300" or<br>$1.45 \times$ "6300" or<br>$145 \times$ "63" $(= 9135)$ | | | | M1 for a method to find Bahira's savings<br>or<br>for 9135 |
| | "9135" $\div \frac{3}{2}$ oe or "9135" $\times \frac{2}{3}$ oe | | | | M1 for a method to find to find Chanda's savings |
| | *Working not required, so correct answer scores full marks (unless from obvious incorrect working)* | | 6090 | | A1 |
| | | | | | **Total 5 marks** |

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